

DecodeLabs AI Internship – Project 1 Report

Project Title

Rule-Based AI Chatbot using Streamlit

Intern Name

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Technology Used

- Python
 - Streamlit
 - Datetime Module
 - Session State
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Project Overview

This project is a Rule-Based AI Chatbot developed using Python and Streamlit. The chatbot interacts with users by responding to predefined commands using conditional (if-elif-else) logic.

Unlike advanced AI models that learn from data, this chatbot follows fixed rules and generates responses based on user input. The purpose of this project is to understand the fundamentals of Artificial Intelligence, decision-making logic, and chatbot development.

Objective

The main objective of this project is to:

- Create a chatbot that can communicate with users.
 - Implement rule-based decision making.
 - Handle predefined commands and responses.
 - Build an interactive user interface.
 - Maintain conversation history during runtime.
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Features Implemented

1. Modern User Interface

A clean and professional chatbot interface was developed using Streamlit.

2. Chat History

The chatbot stores previous messages using Streamlit Session State and displays the complete conversation.

3. User and Bot Avatars

Separate avatars are used for:

- User 
- Bot 

This makes the conversation visually appealing.

4. Timestamp Support

Every message contains a timestamp showing when the message was sent.

5. Sidebar Navigation

A sidebar is included containing:

- Project Information
- Available Commands
- Chat Management Options

6. Clear Chat Function

Users can clear the entire conversation using the "Clear Chat" button.

7. Rule-Based Responses

The chatbot responds to predefined commands using if-elif-else conditions.

Supported Commands:

- hi
- hello

- hey
- how are you
- your name
- time
- date
- help
- bye
- exit

8. Unknown Command Handling

If a user enters an unsupported command, the chatbot displays an appropriate error message and suggests using the help command.

Working of the Chatbot

The chatbot follows the following workflow:

1. User enters a message.
 2. The message is converted to lowercase.
 3. The system checks the input against predefined rules.
 4. A matching response is generated.
 5. Both user and bot messages are stored in Session State.
 6. Messages are displayed with timestamps and avatars.
 7. The chat continues until the user closes the application.
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Program Logic

The chatbot uses conditional statements:

- if
- elif
- else

Example:

If the user types:

"hi"

The chatbot replies:

"Hello! How can I help you today?"

If the user types:

"time"

The chatbot displays the current system time.

If the user types an unknown command, the chatbot displays a fallback response.

How to Run the Project

Step 1: Install Python

Download and install Python from:

<https://www.python.org>

Step 2: Install Streamlit

Open Terminal or Command Prompt and run:

```
pip install streamlit
```

Step 3: Navigate to Project Folder

Open terminal in the folder containing app.py.

Example:

```
cd AI-Chatbot
```

Step 4: Run the Application

Execute:

```
streamlit run app.py
```

Step 5: Open in Browser

Streamlit will automatically open the application in your browser.

Default URL:

<http://localhost:8501>

Learning Outcomes

Through this project, the following concepts were learned:

- Rule-Based Artificial Intelligence
 - Python Programming
 - Conditional Statements
 - Functions
 - Session State Management
 - Streamlit Framework
 - User Interface Design
 - Chatbot Development Fundamentals
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Conclusion

This project successfully demonstrates the implementation of a Rule-Based AI Chatbot using Python and Streamlit. The chatbot provides a user-friendly interface, supports multiple commands, maintains conversation history, and showcases fundamental AI concepts through decision-making logic. The project serves as a strong foundation for developing more advanced AI-powered conversational systems in the future.